Ethereum 2.0

What is Ethereum?

- Ethereum is a network of computers all over the world that follow a set of rules called the Ethereum protocol.
- A decentralized platform that enables smart contracts and decentralized applications (dApps).
- Ether (ETH) is the native cryptocurrency used for transactions and computation fees.
- Created by Vitalik Buterin in 2015, it goes beyond cryptocurrency, providing a platform for developers.

What Can Ethereum Do?

- ➔ **Banking for everyone**
- ➔ **An open internet**
- ➔ **A peer-to-peer network**
- ➔ **Censorship-resistant**
- ➔ **Commerce guarantees**
- ➔ **Composable products**

ETHEREUM

Ethereum in numbers

4K+	96M+	53.3M+
Projects build on Ethereum ⓘ	Accounts (wallets) with an ETH balance ⓘ	Smart contracts on Ethereum ⓘ
$410B	$3.5B	18.82M
Value secured on Ethereum ⓘ	Creator earnings on Ethereum in 2021 ⓘ	Number of transactions today ⓘ

CRYPTOCURRENCY

CRYPTOCURRENCY:

- Cryptocurrencies like Bitcoin and Ether rely on cryptography for security, not on trust in institutions or corporations.
- Ethereum's native cryptocurrency is **Ether (ETH)**:
  - Used to pay for network activities.
  - Transferable between users or exchangeable for other tokens on Ethereum.
- Ether is unique because it funds the computation required to:
  - Build decentralized apps (dApps).
  - Operate organizations on Ethereum.

BLOCKCHAIN

What is blockchain?

- A blockchain is a shared database of transactions distributed across a network of computers.
- Transactions are grouped into "blocks" that are added sequentially to the chain.
- Public blockchains like Ethereum:
  - Allow anyone to add data but not remove it.
  - Ensure transparency and immutability.

BLOCKCHAIN

- Altering information requires changing the majority of computers on the network, which is extremely challenging.
- This decentralized nature makes Ethereum and similar blockchains highly secure.

SMART CONTRACTS

WHAT ARE SMART CONTRACTS?

- Self-executing contracts with the terms of the agreement directly written into code.
- Run on Ethereum’s blockchain, removing the need for intermediaries.
- **Applications:**
  - Automates complex processes, used in DeFi, supply chain, etc.

Consensus Protocols

WHAT IS CONCENSUS PROTOCOL?

Mechanisms used to achieve agreement on the blockchain about the current state of the ledger.

Types:

- Proof of Work (PoW)
- Proof of Stake (PoS)
- Hybrid consensus like Casper and Gasper.

PROOF OF WORK

PROOF OF WORK (POW)

How it Works:

- Miners compete to solve complex mathematical puzzles to add new blocks to the blockchain.
- The first one to solve the puzzle gets to append the block and is rewarded with ETH.

PROOF OF WORK (POW)

Drawbacks:

Energy-intensive and slower finality times.

PROOF OF STAKE

PROOF OF STAKE (POS)

How it Works:

- Validators deposit ETH to be selected to propose blocks and validate transactions.
- Validators are selected to propose and attest to blocks based on the amount of cryptocurrency they "stake."
- More stake = more chances to be selected, incentivizing honest behavior
- Validators are incentivized to act honestly because misbehavior results in slashing of their staked ETH.

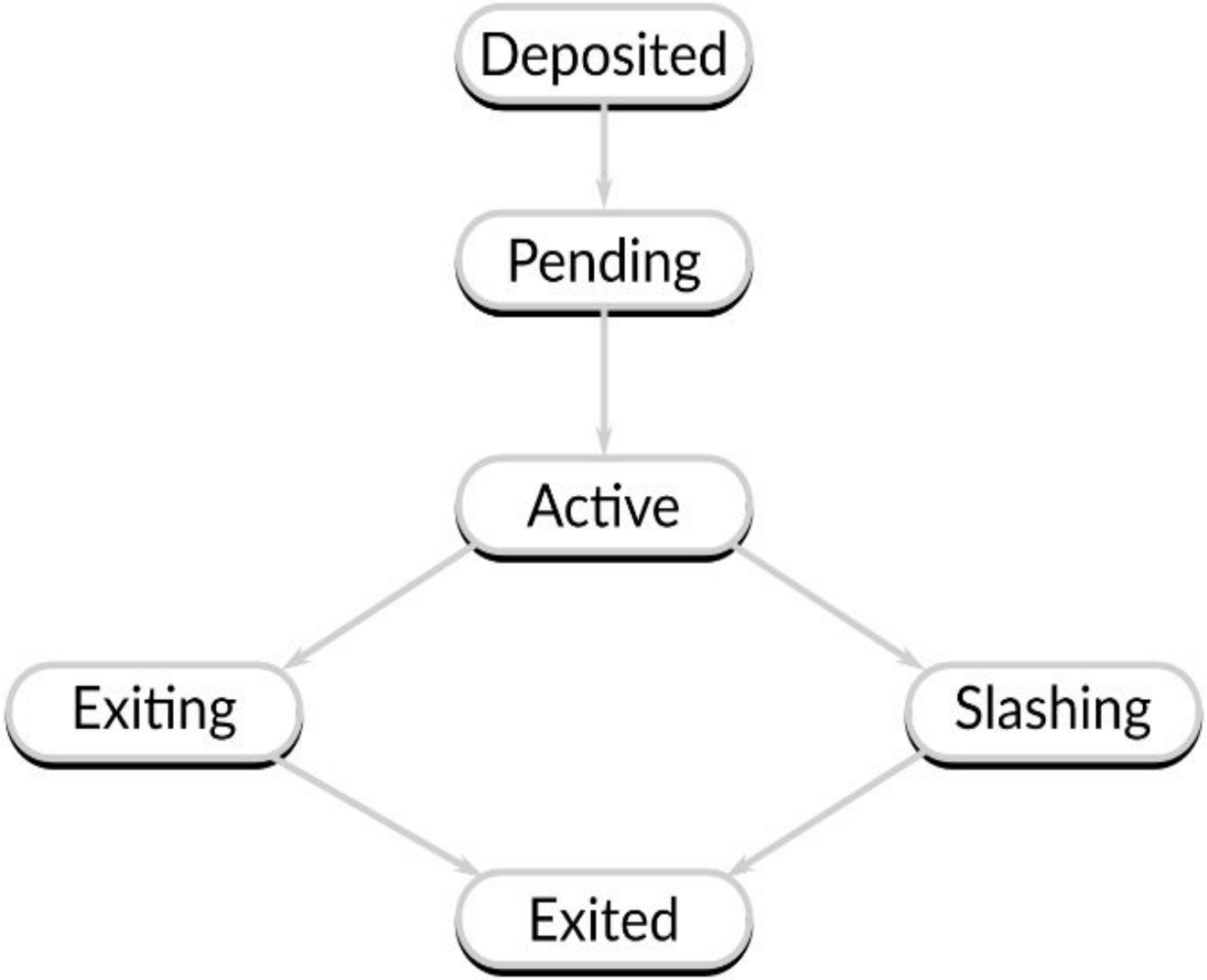
PROOF OF STAKE (POS)

PoS is more energy-efficient and scalable compared to PoW.

VALIDATORS AND STAKE

1. Validators play a crucial role in PoS. They are selected to propose blocks based on their stake, and they also vote on the validity of blocks proposed by others (attestation).
2. Validators risk losing part of their stake (slashing) if they attempt dishonest activities like proposing conflicting blocks

Complete Validator Lifecycle of the Ethereum Beacon Chain


Deposited
Pending
Active
Exiting
Slashing
Exited

VALIDATOR STATUSES IN PHASE 0

- **Deposited** - a deposit agreement has been accepted by the Ethereum 1 network
- **Pending** - the deposit agreement has been accepted by the Ethereum 2 network
- **Active** - the validator is attesting to and proposing blocks
- **Exiting** - the validator is in the process of halting attesting to and proposing blocks
- **Slashing** - the validator is in the process of halting attesting to and proposing blocks, having been caught cheating
- **Exited** - the validator is no longer attesting

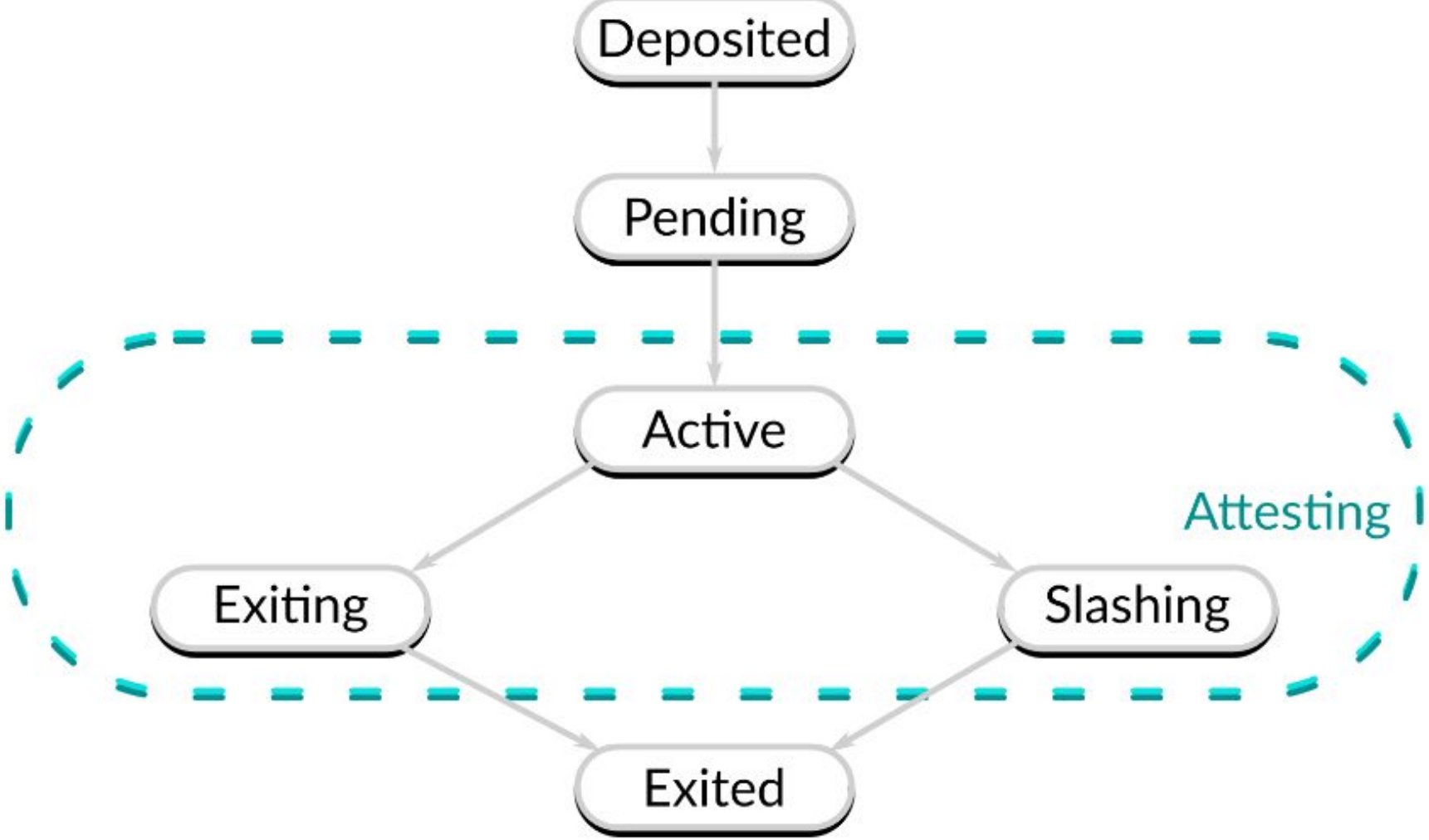


Figure 2: The attesting meta-state

Deposited

1. User submits 32 ETH to the validator deposit contract on Ethereum 1.0.
2. The validator will remain in the deposited state for approximately 7 and a half hours2, to ensure that there is no realistic chance of a chain reorganization affecting the Ethereum 1 transaction and potentially invalidating the deposit agreement.



Figure 3: Transition from deposited to pending

PENDING (Eligible for Activation)

1. Deposit is confirmed and processed by the **Beacon Chain**.
2. Validator becomes **eligible for activation**.
3. Deposits are processed in **batches**.
4. **Activation delay** may occur for security recognition.

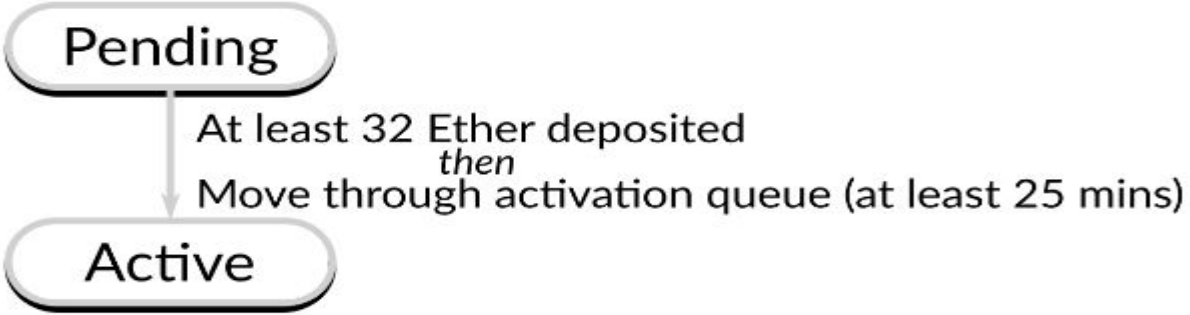


Figure 4: Transition from pending to active

ACTIVE

1. Validator **participates** in consensus by proposing and attesting to blocks.
2. Assigned specific **duties each epoch** (32 slots, 12 seconds per slot).
3. Validators **receive rewards** for active participation.
4. Transition from pending to active involves a **queuing process**.

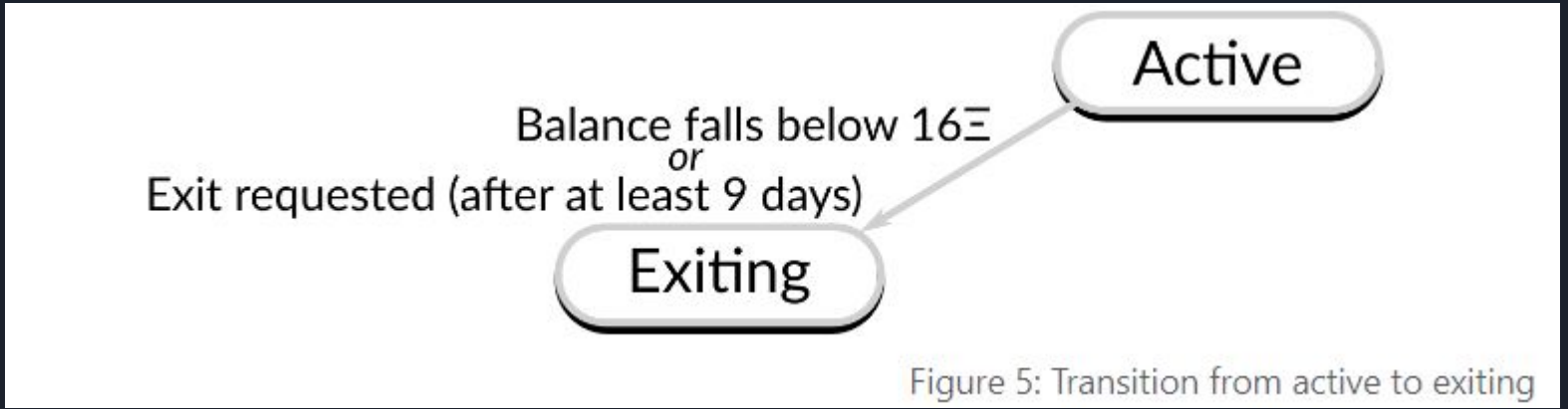


Figure 5: Transition from active to exiting

EXITING

1. A validator may exit **voluntarily** or due to **penalties**.
2. **Voluntary exits** occur when a validator chooses to stop participating.
3. **Involuntary exits** result from actions like double voting.
4. Duties are **reduced** during the exit process, and penalties may apply.

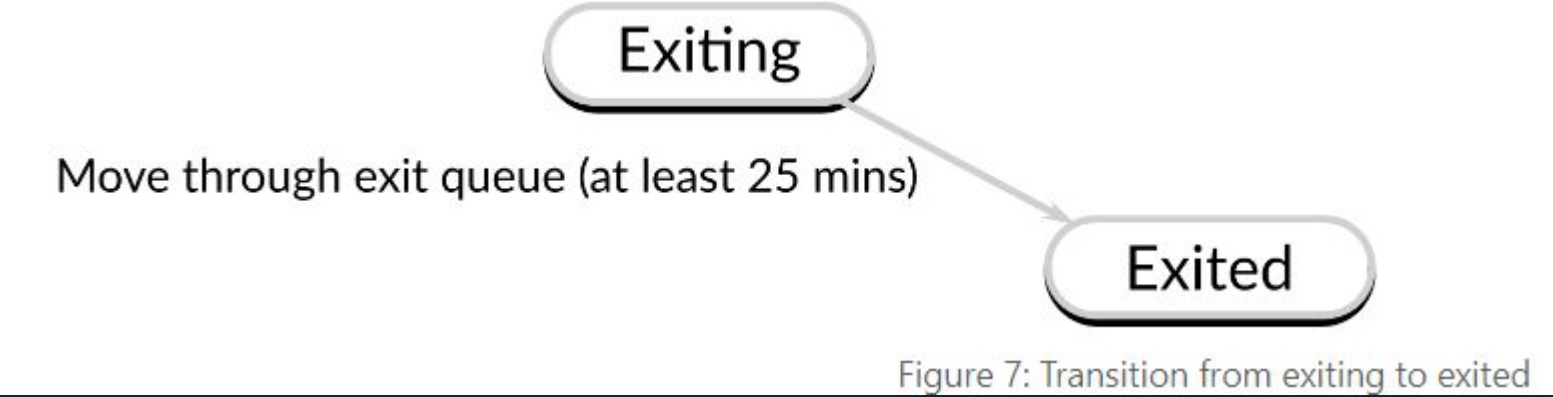


Figure 7: Transition from exiting to exited

SLASHED

1. Validators found to have **misbehaved** (e.g., double voting) are slashed.
2. **Penalties** reduce the staked balance.
3. Slashed validators face a **withdrawal delay** before exiting.

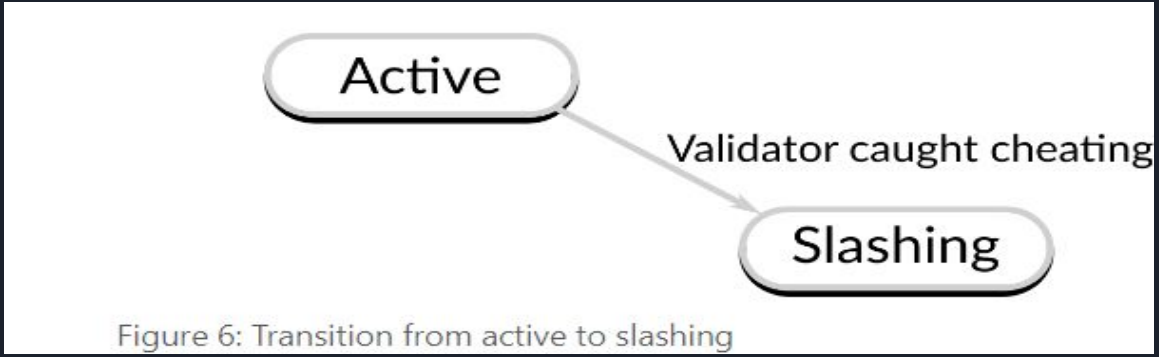


Figure 6: Transition from active to slashing

EXITED

1. Validator completes the **exit process** and is no longer active.
2. Can withdraw the remaining stake after a **specified period**.
3. Withdrawal eligibility depends on exit type (voluntary or slashed).

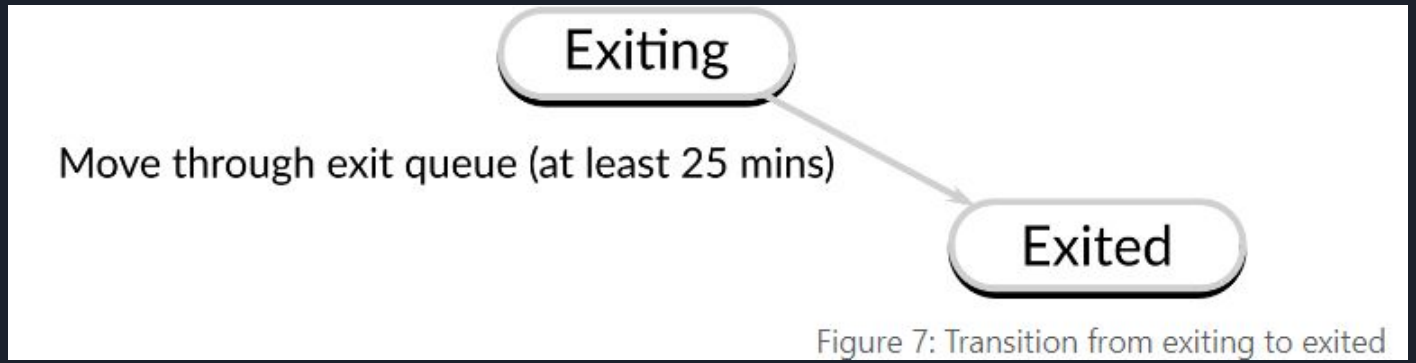
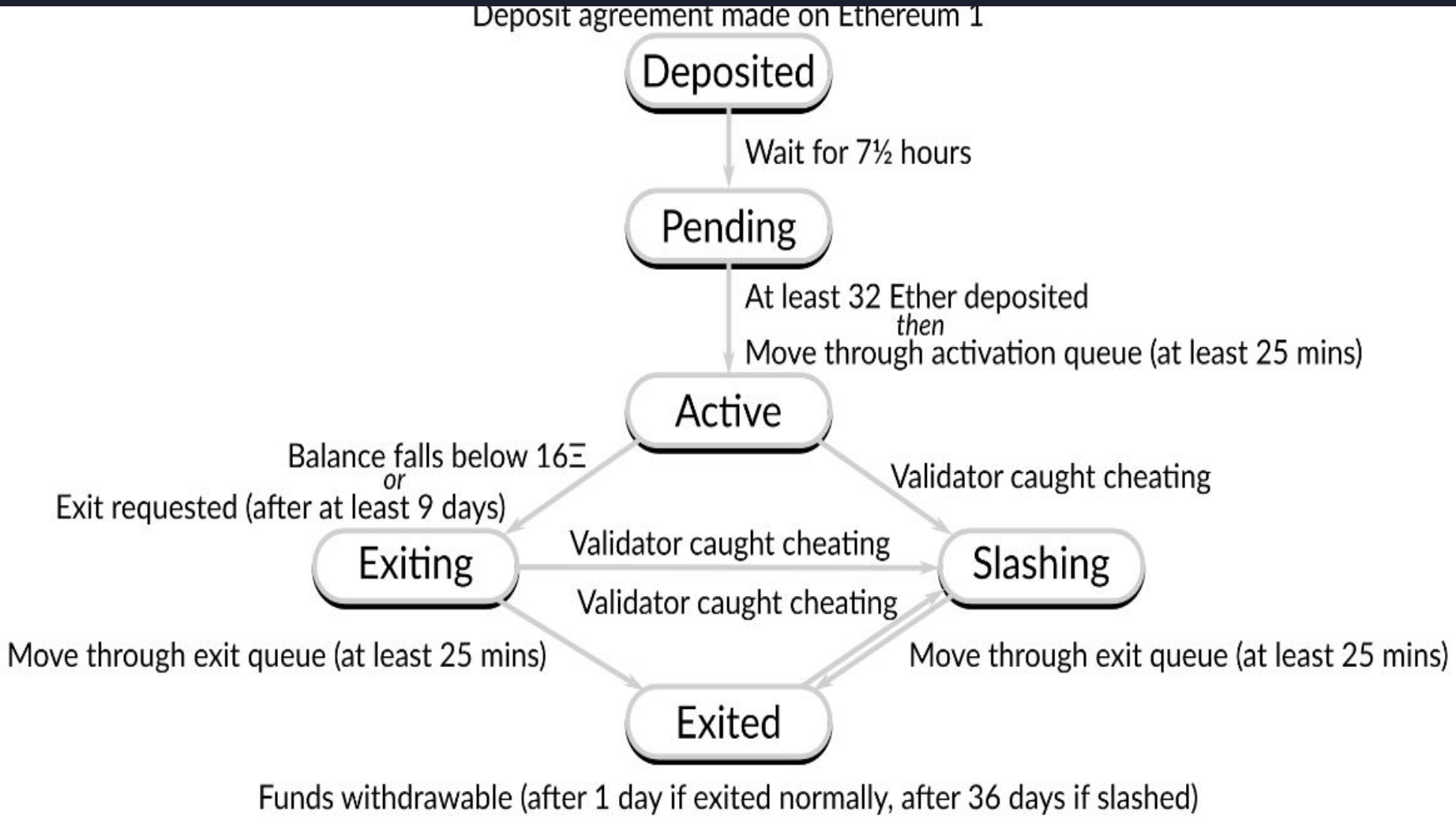


Figure 7: Transition from exiting to exited

WITHDRAWABLE

1. After exit, any applicable penalties are applied.
2. Validator's remaining stake becomes **withdrawable**.
3. **Withdrawal eligibility** time varies; slashed validators face longer delays.

EXTENDED VALIDATOR LIFE CYCLE


Deposit agreement made on Ethereum 1
Deposited
Wait for 7½ hours
Pending
At least 32 Ether deposited
then
Move through activation queue (at least 25 mins)
Active
Balance falls below 16Ξ
or
Exit requested (after at least 9 days)
Validator caught cheating
Exiting
Validator caught cheating
Slashing
Validator caught cheating
Move through exit queue (at least 25 mins)
Move through exit queue (at least 25 mins)
Exited
Funds withdrawable (after 1 day if exited normally, after 36 days if slashed)

EPOCS AND SLOTS

EPOCHS AND SLOTS

- **SLOT:**

  A 12-second time window for a committee of Validators to complete their actions.

- EPOCH:

  Epochs consist of 32 slots, and each slot is an opportunity for a block to be proposed.

Validators are assigned specific slots to propose blocks. If no block is proposed in a slot, it remains empty.

BLOCK PROPOSAL PROCESS

BLOCK PROPOSAL PROCESS

Random Selection: Block proposers are selected randomly in PoS.

Beacon Chain Shuffle:

- At the start of each epoch, Ethereum performs a beacon chain shuffle.
- This shuffle assigns validators to specific tasks, including block proposals.

Lookahead Function:

- The shuffle provides validator assignments one epoch in advance.
- This lookahead allows validators to prepare for their assigned tasks.

ATTESTATION

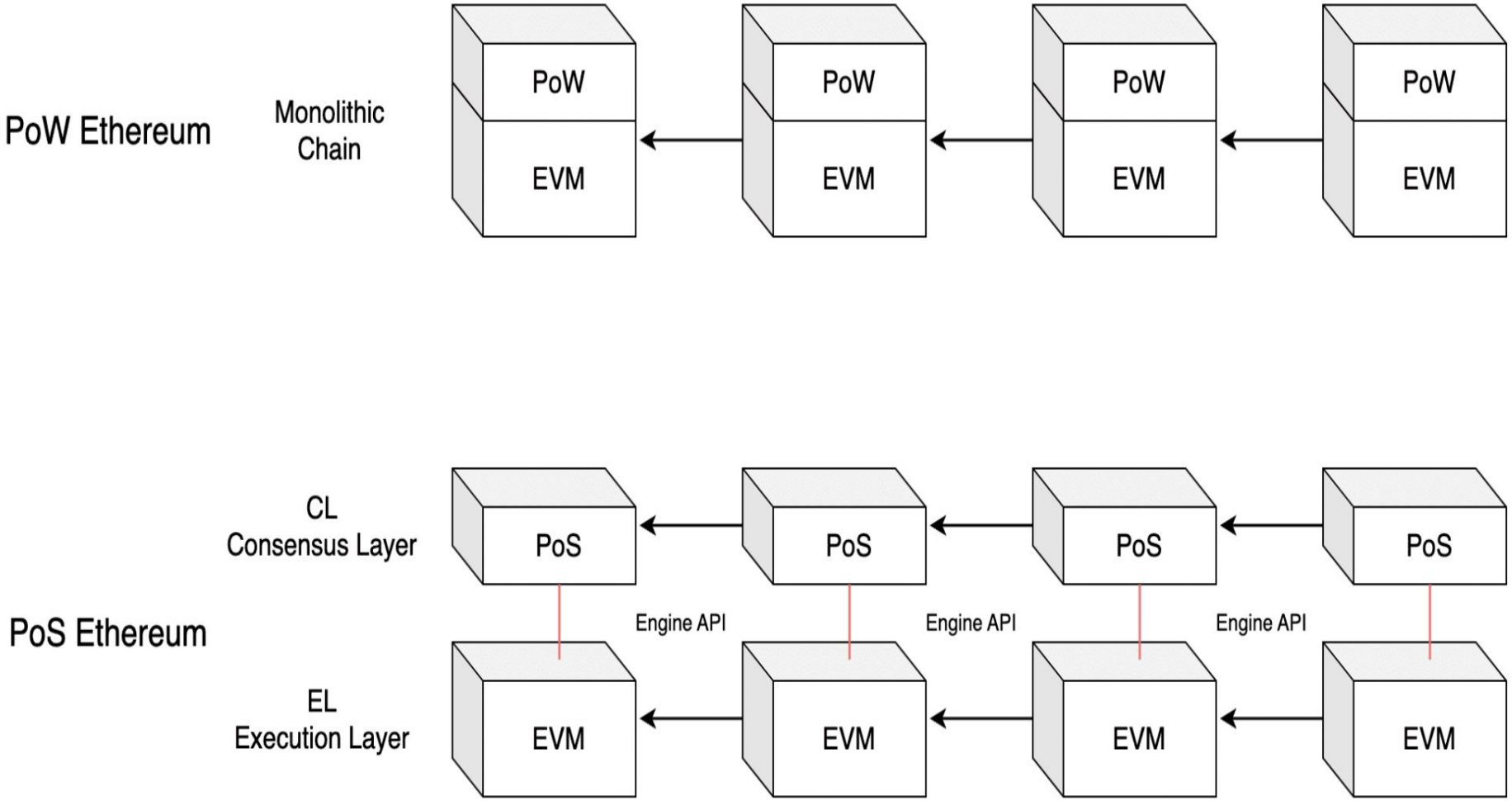
ATTESTATION

1. Validators “attest” (vote) on the validity of a block.
2. Attestations are critical for ensuring that the chain grows in a secure and consistent manner.
3. These attestations are included in future blocks as evidence of the chain’s correctness.

FORK CHOICE RULE

WHAT IS FORK CHOICE?

A set of rules determining which chain to follow in the case of multiple chains.


PoW Ethereum
Monolithic Chain
PoW
EVM
PoW
EVM
PoW
EVM
PoW
EVM
PoS Ethereum
CL Consensus Layer
PoS
PoS
PoS
PoS
Engine API
Engine API
Engine API
EL Execution Layer
EVM
EVM
EVM
EVM

ETHEREUM'S LATEST FORK CHOICE

- LMD-GHOST (Latest Message Driven Greediest Heaviest Observed Subtree), where validators follow the heaviest subtree of votes.

CASPER FFG (FINALITY GADGET)

WHAT IS CASPER?

1. Ethereum Casper is a protocol that combines both POW and POS.
2. The Proof of Work part uses the Ethash Algorithm.
3. Validators replace miners in PoS, staking their ETH to propose and validate blocks.
4. It introduces a checkpointing system where validators vote on blocks to justify and finalize them.

GASPER PROTOCOL

WHAT IS GASPER?

1. **Gasper** combines Casper FFG with LMD-GHOST. Casper handles finality, ensuring that justified blocks become irreversible, while LMD-GHOST ensures that the chain with the most votes (attestations) is selected.
2. **Liveliness:** LMD-GHOST ensures the chain keeps progressing (transactions continue happening).
3. **Security:** Casper guarantees that once a block is finalized, it cannot be reverted, making the blockchain tamper-proof.

JUSTIFICATION AND FINALIZATION

JUSTIFICATION

1. A checkpoint in the blockchain is justified when it has sufficient validator votes.
2. A checkpoint block is justified if it receives a supermajority (2/3) of validator votes.

FINALIZATION

1. A justified block becomes final when it is followed by another justified block.
2. Finality ensures that a block cannot be reverted, bringing security to Ethereum’s PoS chain.

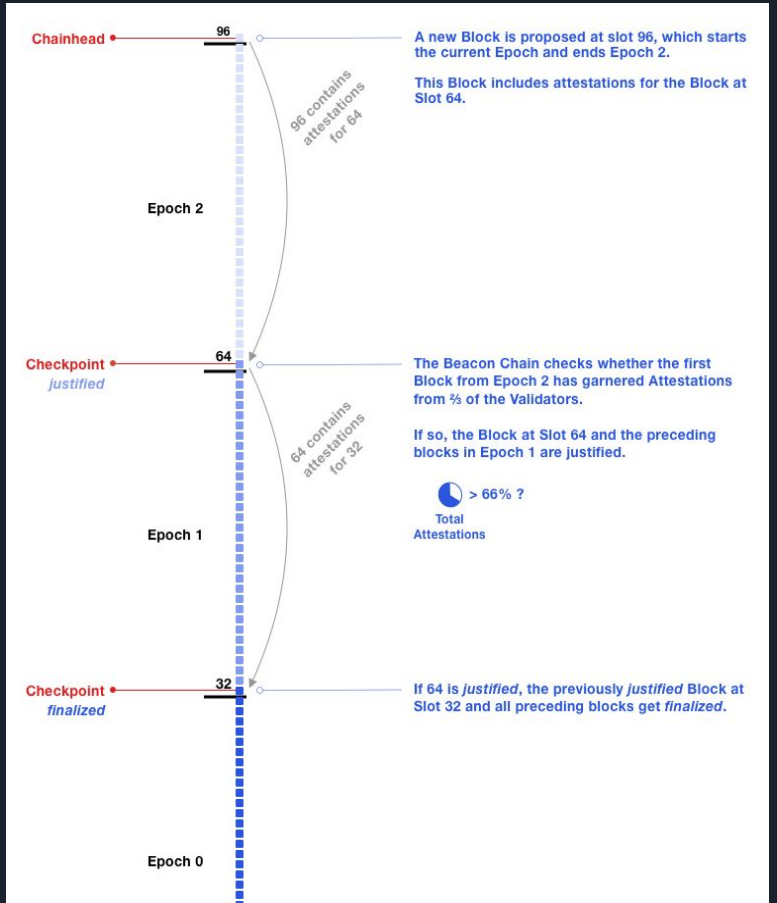
JUSTIFICATION AND FINALIZATION

Slot 96 (Chainhead):

- A new block is proposed at Slot 96, marking the start of the current epoch (Epoch 2) and the end of the previous one (Epoch 1).
- This block contains attestations for the block proposed at Slot 64.

Slot 64 (Checkpoint - Justified):

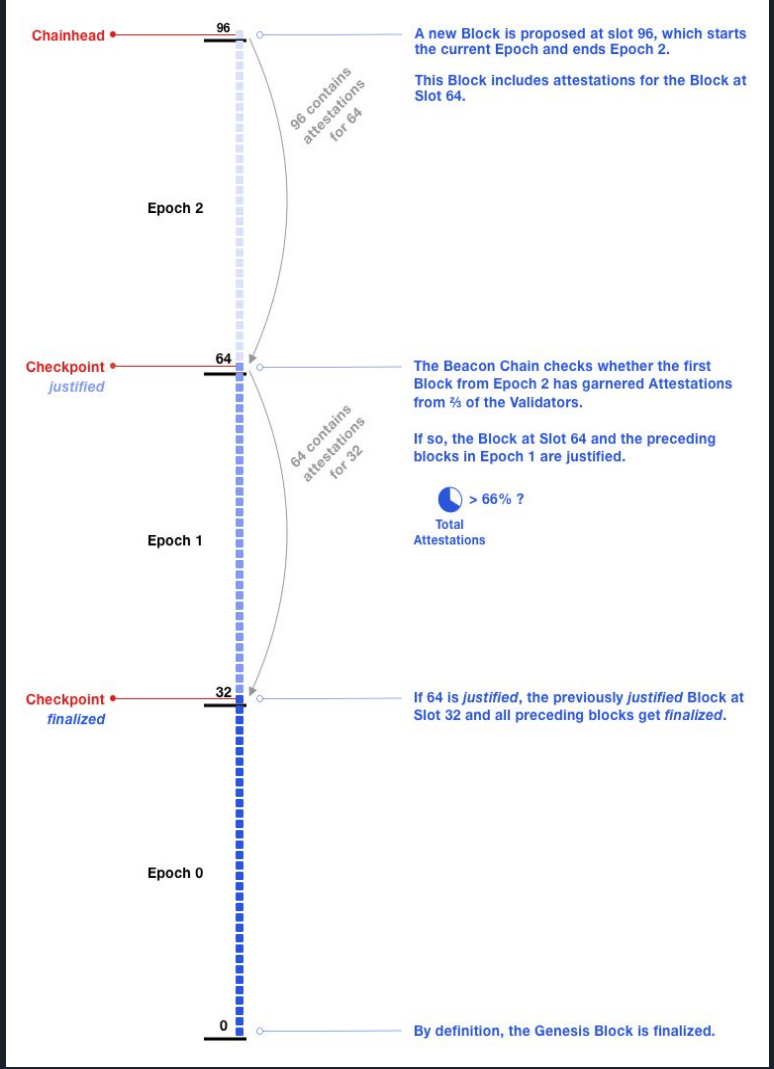
- The Beacon Chain checks if the first block from Epoch 2 (Slot 64) has gathered attestations from at least **2/3 of the validators**.
- If it has, the block at Slot 64, along with the preceding blocks in Epoch 1, is considered **justified**.



JUSTIFICATION AND FINALIZATION

Justification Process:

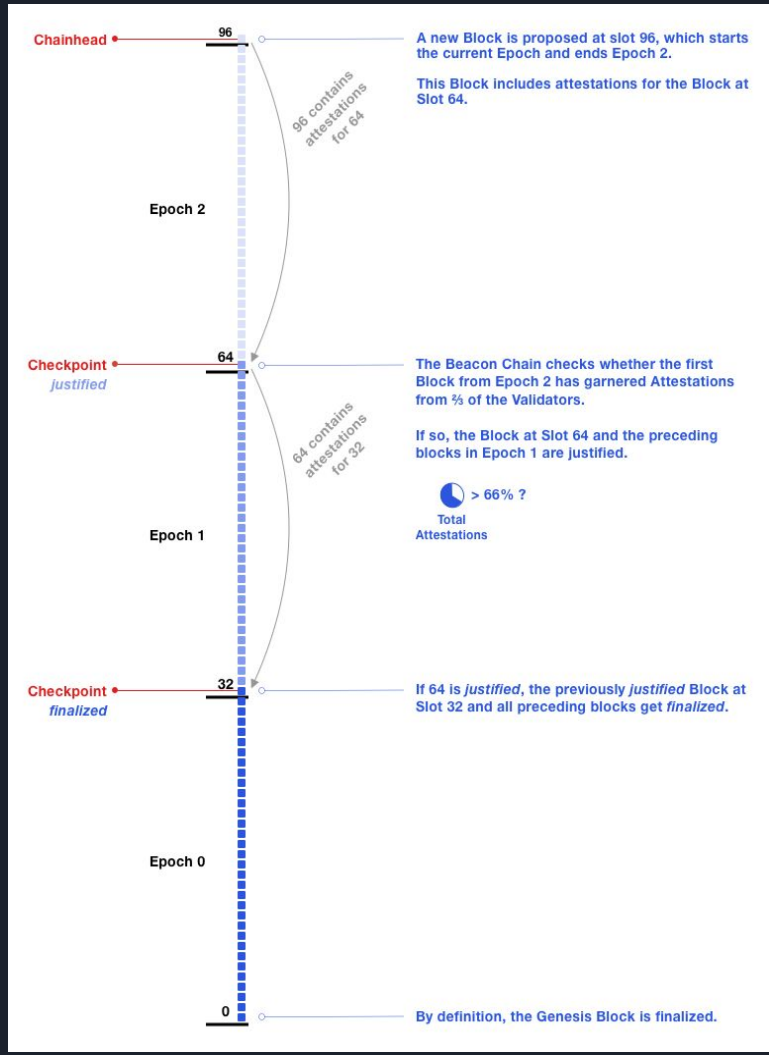
- Justification means that the block at Slot 64 has received sufficient attestations from validators, indicating consensus.
- This process helps secure the blockchain and builds confidence in earlier blocks.



JUSTIFICATION AND FINALIZATION

Slot 32 (Checkpoint - Finalized):

- If Slot 64 is justified, then the previously justified block at Slot 32 (and all preceding blocks, including the Genesis block) becomes **finalized**.
- Finalization is an important part of consensus, as it means blocks can no longer be reverted.



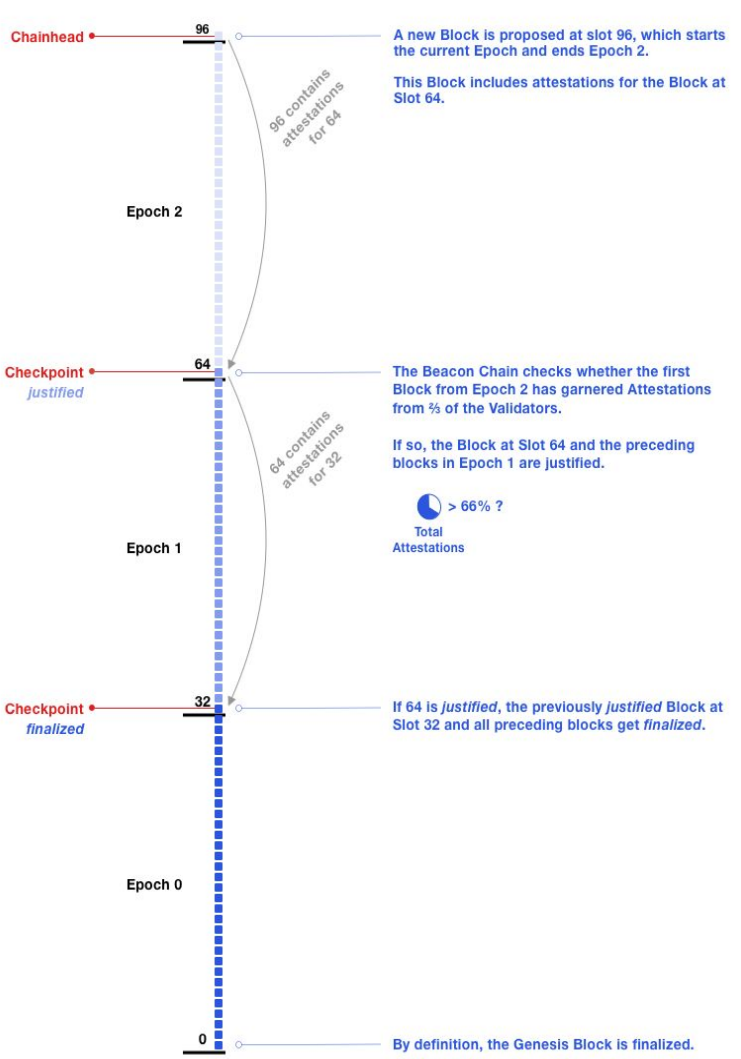
JUSTIFICATION AND FINALIZATION

Finalization of Genesis Block:

- By definition, the Genesis block at Slot 0 is **finalized**.

Threshold for Finalization:

- For blocks to be finalized, more than **66% of the total attestations** must confirm the justification of a block.



GAS IN ETHEREUM

GAS IN ETHEREUM

1. Gas is a unit measuring the computational work required to execute a transaction or smart contract.
2. The user sets a gas price and gas limit, determining the total transaction fee paid in Ether.

Ethereum’s Scalability Challenges and Sharding

Sharding is an upgrade in Ethereum aimed at improving scalability by splitting the network into smaller, manageable pieces (shards). Each shard processes transactions separately, improving throughput and reducing congestion.

CONCLUSION

CONCLUSION:

Ethereum’s PoS-based system, enhanced by Casper and Gasper, ensures scalability, energy efficiency, and robust security through justification and finalization.